



# Long- and short-term antibacterial properties of low-density polyethylene-based films coated with zinc oxide nanoparticles for potential use in food packaging

Journal of Plastic Film &amp; Sheeting

0(0) 1–18

© The Author(s) 2019

Article reuse guidelines:

[sagepub.com/journals-permissions](http://sagepub.com/journals-permissions)

DOI: 10.1177/8756087918822677

[journals.sagepub.com/home/jpf](http://journals.sagepub.com/home/jpf)

Hajer Rokbani<sup>1</sup> , France Daigle<sup>2</sup> and  
Abdellah Ajji<sup>1</sup>

## Abstract

Concerns in food safety and the need for high-quality foods have increased the demand for extending the shelf life of packaged foods. Subsequently, promoting and investigating the development of antibacterial materials for food packaging has become inevitable. Zinc oxide nanoparticles have attracted attention lately owing to their multifunctional properties, especially antibacterial activity. For this study, antibacterial low-density polyethylene films were prepared by coating zinc oxide nanoparticles onto their surface. The low-density polyethylene film antibacterial activity was evaluated toward Gram-positive and Gram-negative bacteria. The scanning electron microscopy images showed that using anhydride-modified low-density polyethylene (LDPE-g-AM) resin permitted improved zinc oxide nanoparticle distribution on the low-density polyethylene film surface, reduced the agglomerate sizes, and reinforced the zinc oxide nanoparticle bonding to the low-density polyethylene film surface. We found that the coated

<sup>1</sup>3SPack, CREPEC, Department of Chemical Engineering, Polytechnique Montréal, Montreal, Canada

<sup>2</sup>Department of Microbiology, Infectiology and Immunology, Pavillon Roger-Gaudry, Université de Montréal, Montréal, Canada

## Corresponding author:

Abdellah Ajji, 3SPack, CREPEC, Department of Chemical Engineering, Polytechnique Montréal, P.O. Box 6079, Station Centre-Ville, Montreal, Quebec, Canada H3C 3A7.

Email: [abdellah.ajji@polymtl.ca](mailto:abdellah.ajji@polymtl.ca)

low-density polyethylene films exhibited high antibacterial activity against both strains. The antibacterial tests also proved that the coated films retained their antibacterial efficiency toward *Escherichia coli*, even after eight months, with a reduction rate higher than 99.9%, whereas for *Staphylococcus aureus* the antibacterial properties for the linear low-density polyethylene (LLDPE) films decreased at eight months and improved for the LDPE-g-AM films. When the zinc oxide coated films were laminated with neat low-density polyethylene, only the LDPE-g-AM was still active against *E. coli* provided that the lamination thickness does not go beyond 8  $\mu\text{m}$ . This research demonstrated that the coated low-density polyethylene films have excellent attributes when used as an active coating in the food packaging industry.

### Keywords

Zinc oxide nanoparticles, low-density polyethylene, food packaging, antimicrobial activity, coating

## Introduction

Recent innovations in the food packaging industry have transformed passive packaging, which was simply a container or insulating barrier for food and beverage from the outer environment into a more active role in the packaging industry.<sup>1</sup> As such, active packaging can prevent moisture gains or losses, microbial contamination, and be a barrier against gases such as oxygen, carbon dioxide, water vapor, and other compounds such as flavor.<sup>2</sup> Correspondingly, in recent years, food-borne diseases are a concern for food packaging, distribution, and storage. Even though techniques in sanitation and the inspection of food production have evolved, microorganisms that cause food spoilage and carry diseases are risks during food processing, packaging, shipping, storage, and distribution.<sup>3</sup> Antibacterial (AB)-active films are an innovative concept in food packaging developed in response to satisfy consumer expectations for safe food free from contamination.<sup>4</sup> In fact, AB food packaging systems are polymeric materials in which additives with pronounced AB properties are embedded into the plastic packaging material (either rigid or flexible packaging) to extend the food shelf life,<sup>5</sup> by controlling pathogenic and spoilage microorganism growth.

Advances in nanotechnology in the food packaging industry is aimed to provide solutions to the challenges in the food packaging industry addressing problems such as a short shelf life and better performances in AB packaging.<sup>6</sup> Metal oxides like titanium oxide ( $\text{TiO}_2$ ), zinc oxide ( $\text{ZnO}$ ), calcium oxide ( $\text{CaO}$ ), and magnesium oxide ( $\text{MgO}$ ) including metals such as silver and gold have been developed for the preparation of AB packaging films due to their strong AB characteristics, high stability, robustness, and long shelf life compared with organic AB agents.<sup>2</sup> Their AB efficiency has been widely investigated by many researchers.<sup>7-9</sup> Silver is

commonly used in AB nanocomposites for food packaging and is well known for its strong toxicity against many microorganisms.<sup>10</sup> However, zinc oxide nanoparticles (ZnO-NPs) can compete favorably with silver nanoparticles (NPs), especially because of their simple and inexpensive synthesis, greater efficiency, cost-effectiveness, lower levels of toxicity,<sup>2,11</sup> and their toxicity which is selective toward a broad range of both Gram-positive and Gram-negative bacteria, including harmful food-borne organisms such as *Escherichia coli*, *Listeria* and *Staphylococcus aureus*.<sup>12–15</sup> Moreover, ZnO-NPs show an antibacterial activity (ABA) without photo-activation compared to TiO<sub>2</sub>.<sup>12,15,16</sup>

ZnO is used in daily applications, for example, in drug delivery, cosmetics, and medical industries due to its potent AB effect against many microorganisms.<sup>17</sup> Moreover, the Food and Drug Administration currently lists ZnO as a generally recognized as safe material. The ZnO-NP ABA depends on concentration and particle size,<sup>15</sup> with shape and structure being inconsequential.<sup>18</sup> Higher concentrations and larger surface areas have better activity.<sup>14</sup> When incorporated into an AB package, the ZnO-NPs improve mechanical strength, barrier properties, and stability.<sup>13,19</sup> As the most commonly used food packaging materials are natural and synthetic polymers,<sup>20</sup> and because they have better physical and barrier properties, higher strength and elongation, lower cost and lightness, plastic films play an important role in food distribution.<sup>21</sup> ZnO-NPs have been integrated into various material using different methods including polylactide,<sup>22,23</sup> glass,<sup>24</sup> low-density polyethylene (LDPE),<sup>6,25,26</sup> polypropylene,<sup>23,27</sup> polyurethane,<sup>28</sup> poly(vinyl chloride) (PVC),<sup>18</sup> poly(butylene succinate),<sup>29</sup> polyethylene terephthalate (PET),<sup>30</sup> paper,<sup>31</sup> and chitosan.<sup>32,33</sup> LDPE is primarily used as it possesses desirable properties such as good flexibility, transparency, economical cost efficiency, ability to be processed easily, and thermal stability.<sup>6</sup> How ZnO-coated films are prepared depends on the system being developed. Basically, the AB system can be either in direct or indirect contact. If direct contact is needed, the AB system should be a non-migratory system. Thus, dry or thermo-mechanical techniques including standard conventional techniques used in the plastics industry such as extrusion, compression molding, blow molding, and injection molding should be used. With a volatile releasing system, the coating method can be used where an AB agent with a high concentration is applied to the polymeric film surface.<sup>34</sup> Some commonly accepted AB mechanisms of ZnO-NPs is forming a reactive oxygen species (ROS),<sup>12,35–37</sup> interacting the NPs with bacteria<sup>14,38–40</sup> and the releasing Zn<sup>2+</sup> ions.<sup>41,42</sup> The generated ROS should increase with the ZnO-NP concentration,<sup>12,42</sup> they are expected to oxidize the lipid membrane of the cell wall of *E. coli*. The direct interaction between ZnO-NPs and the cell surface affects the membrane permeability,<sup>14,40</sup> allows the internalization of NPs<sup>38</sup> and induces oxidative stress in bacteria cell.<sup>43</sup> The Zn<sup>2+</sup> ion solubility in the micro-organisms medium could cause the ZnO-NP toxicity.<sup>13,19</sup> Research has shown that ZnO-NPs possess better activity toward *E. coli* as Gram-negative bacteria than toward *S. aureus* as Gram-positive bacteria. The high *S. aureus* resistance to ZnO-NPs can be interpreted to

the structural differences in the bacterial membranes and the intracellular antioxidant content of *S. aureus*.<sup>44</sup>

This study explores producing AB-LDPE films by coating with ZnO-NPs in the last step of the extrusion process. How LDPE-g-AM enhances the ZnO-NP distribution on the polymeric film surface and decreases their agglomeration will be evaluated. To the best of our knowledge, this coating method has never been used to develop AB coated films with ZnO-NPs. The main objectives are to evaluate the AB efficiency of the prepared films and its ability to impede the growth of food-borne bacteria such as *E. coli* and *S. aureus* at different times after films preparation and after being treated in an ultrasonic bath. The distribution and the concentration of the ZnO-NPs on the surface of the coated-films before and after treatment were investigated by EDS-SEM (energy-dispersive X-ray spectroscopy-scanning electron microscopy). Finally, the ABA was compared to films after thin neat LDPE layer lamination and to ZnO-NPs blended with LDPE via the melt compounding.

## Methodology

### Materials

We used the following polymers:

- Surpass FPs317-A Octene sLLDPE film resin from Nova Chemicals (Calgary, Alberta, Canada), melt flow index = 4 g/10 min.
- Bynel 42E703 adhesive resin from DuPont (Mississauga, Ontario, Canada) which is an LDPE-g-AM resin with melt flow index = 5.7 g/10 min and maleic anhydride (MAH) = 0.08 (wt%).

The LDPE grades used in this study were a food-contact grade. Commercially produced ZnO nanofillers were purchased from SkySpring Nanomaterials Inc. (Houston, TX, USA) and were used without any additional purification where the average diameter is 20 nm and the specific surface area is 30 to 50 m<sup>2</sup>/g. Cultures of non-pathogen *E. coli* (DH5 $\alpha$ ) and pathogen *S. aureus* (54-73), Luria Bertani (LB) broth, which is a nutritional-rich medium, LB agar and phosphate-buffered saline (PBS) solutions, used during the AB tests were obtained from the Department of Microbiology, Infectiology, and Immunology at Université de Montréal.

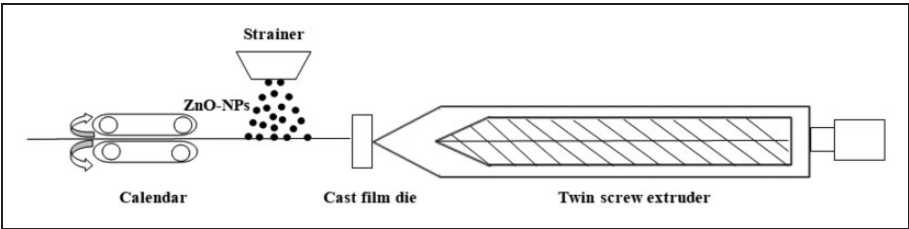
### ZnO-coated LDPE film preparation

LDPE films were prepared by melt blending in a tightly intermeshing co-rotating twin screw extruder (TSE) (Leistritz ZSE-18HP-40D, Germany) with an 18-mm screw diameter and L/D = 40. All the LDPE films were produced at 100 r/min screw speed and a cast film die connected to the TSE. Table 1 shows the temperature profiles.

The ZnO-NPs were manually spread at the die exit with a strainer (Figure 1).

**Table 1.** Different temperature profiles used to produce the LDPE films by the tightly inter-meshing co-rotating twin screw extruder (TSE) (screw diameter = 18 mm and L/D = 40), where T1 = 190, T2 = 200, and T3 = 220°C.

| Zones    | 1 (°C) | 2 (°C) | 3 (°C) | 4 (°C) | 5 (°C) | 6 (°C) | 7 (°C) | 8(°C) | 9 (°C) |
|----------|--------|--------|--------|--------|--------|--------|--------|-------|--------|
| FPs-T1   | 190    | 190    | 190    | 190    | 190    | 190    | 190    | 190   | 190    |
| FPs-T2   | 200    | 200    | 200    | 200    | 200    | 200    | 200    | 200   | 200    |
| FPs-T3   | 200    | 200    | 200    | 200    | 220    | 220    | 220    | 220   | 220    |
| Bynel-T3 | 200    | 200    | 200    | 200    | 220    | 220    | 220    | 220   | 220    |



**Figure 1.** Schematic of the extrusion/coating process.

In fact, FPs-T1, FPs-T2, and FPs-T3 films were processed from FPs317-A resin at 190, 200, and 220°C, respectively. Bynel-T3 films were produced from Bynel 42E703 resin at 220°C. To laminate the coated-LDPE films, the dipping method was used. A 2 wt% FPs317-A xylene solution was prepared at 120°C. The completely dissolved resin solution is cooled to 70°C, and the films were dipped for 1–2 s then dried under the hood overnight.

Characterization

Antibacterial tests

The nanocomposite film AB efficiency was quantified using the strains *E. coli* (DH5α) and *S. aureus* (54-73). These bacteria were grown in LB broth under constant agitation for 24 h at 37°C, to reach a density of 10<sup>9</sup> colony forming units (CFU)/mL. The bacteria culture was diluted in the same medium after 24 h to 10<sup>5</sup> CFU/mL. AB tests were conducted on the films produced by ZnO coating and melt mixing according to the standard ISO 22196:2011 and as described by Abdali and Ajji.<sup>3</sup> Both active and blank samples were cut out into 4- to 5-cm discs. ISO 22196:2011 standards require that the surface of contact between the two polymer layers should be between 400 and 1600 mm<sup>2</sup>.<sup>4,5</sup> Blank discs should be sterilized by washing with 70% ethanol before the test, whereas active films were sterilized under UV light (λ = 254 nm) for 20 min. The two discs (one blank and the other active) are sandwiched together after 200 μL of

$10^5$  CFU/mL culture is deposited between them. Both films are rubbed against each other, so that the surface is contaminated. These are placed in a Petri dish and incubated at  $37^\circ\text{C}$  for 24 h. The next day, both discs are quenched in a test tube with 2 mL PBS solution. Then, for each tube, six serial PBS dilutions ( $10^{-1}$  to  $10^{-6}$ ) were conducted. Three samples were taken from each of the six dilutions and 10  $\mu\text{L}$  droplets were applied onto the LB agar plates which were incubated overnight at  $37^\circ\text{C}$  for counting the surviving bacteria (CFU/mL). Each sample is made in triplicate, along with the negative control containing two blank discs.

To study how well the ZnO-NPs adhere to the film, the ABA of the ZnO-coated LDPE films before and after shaking were investigated. The samples were immersed in a beaker with distilled water and were shaken at  $30^\circ\text{C}$  in an ultrasonic bath (Thermo Fischer, 40 kHz) for 20 min. All the AB tests were conducted in the Department of Microbiology, Infectiology, and Immunology at Université de Montréal.

### Morphology

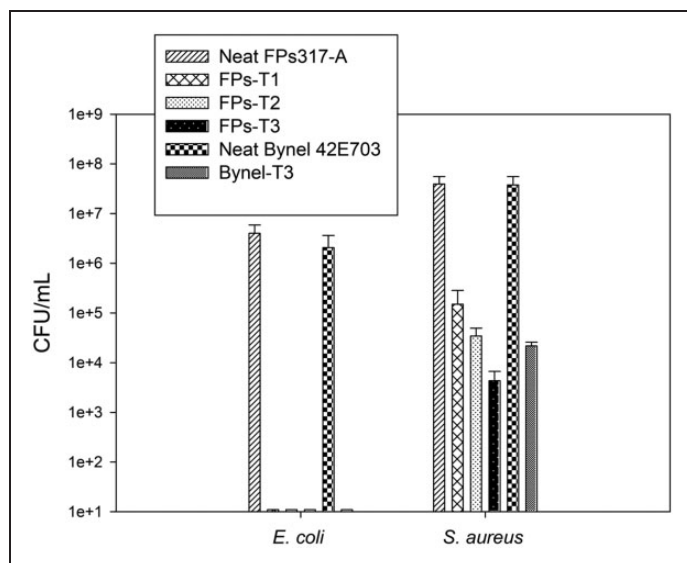
The ZnO-NP distribution on the LDPE film surface was assessed using a field emission gun-scanning electron microscopy JSM-7600TFE (FEG-SEM) (Calgary, Alberta, Canada) operating at 5–10 kV. The surface composition analysis was performed in the same SEM equipment under EDS mode operated at 10 kV.

## Results and discussions

### Coated film AB properties

The ABA was tested against two bacterial strains and was measured by viable cell counts after incubation for 24 h at  $37^\circ\text{C}$  (Figure 2).

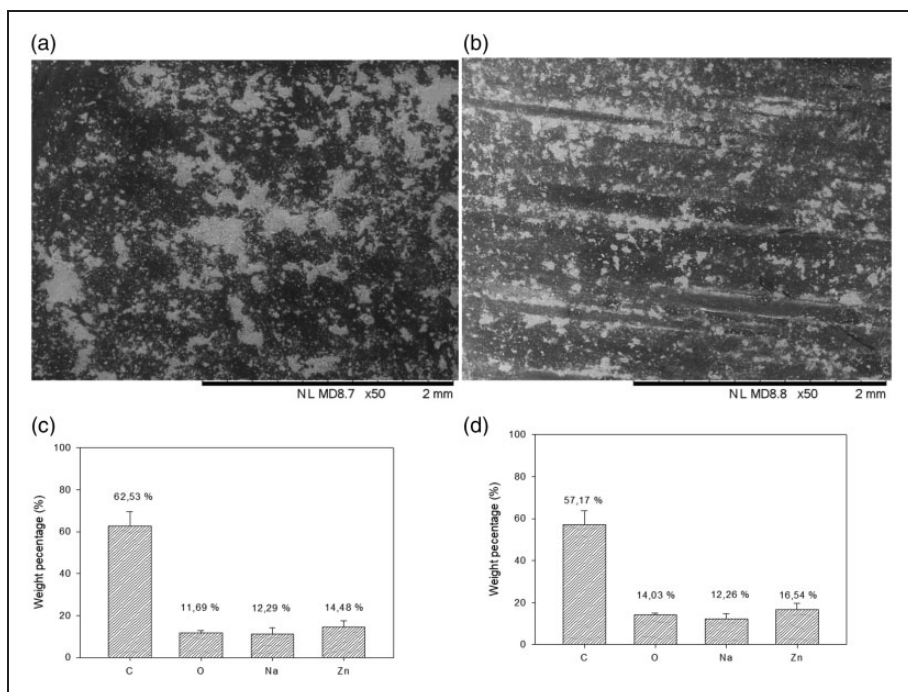
The *E. coli* population was completely eradicated when compared with the control samples and a reduction rate higher than 99.9999%. *S. aureus*, for both FPs and Bynel films, was reduced by more than two logs (CFU/mL) for all samples. These results also proved that increasing the extrusion temperature from  $190^\circ\text{C}$  (FPs-T1) to  $220^\circ\text{C}$  (FPs-T3) improved the AB activity. In fact, the increase in the processing temperature has probably impregnated higher ZnO-NP concentration on the coated LDPE film surface. The higher ZnO-NP concentration on the surface decreased the *S. aureus* population from  $1.5 \times 10^5$  CFU/mL for FPs-T1 films to  $4.3 \times 10^3$  CFU/mL for FPs-T3 films processed at 190 and  $220^\circ\text{C}$ , respectively. This study showed that the ZnO-coated LDPE films exhibited increased ABA on inhibiting the Gram-negative (*E. coli*) growth than the Gram-positive (*S. aureus*) bacteria. Several studies indicate that Gram-positive (*S. aureus*) bacteria have better resistance to ZnO-NPs, than *E. coli* due to the intracellular antioxidant content within *S. aureus* which provides better resistance to oxidation as well as the powerful detoxification agents within the Gram-positive bacteria.<sup>44</sup>



**Figure 2.** Inhibitory effects of different FPs and Bynel films toward *E. coli* and *S. aureus* at one month post-preparation.

To observe and compare the ZnO-NP distribution across the LDPE film surface, FPs-T3 and Bynel-T3 were analyzed by SEM, since these two samples were processed at the same conditions (200–220°C). The SEM micrographs presented in Figure 3(a) and (b) compare the distribution of the ZnO-NP distribution on the surface of the coated film surface.

The photographs clearly show that the ZnO-NP distribution on the Bynel-T3 film surface was more uniform, whereas agglomerations were observed on the FPs-T3 surface, possibly because of the strong interactions between ZnO-NPs. In fact, Bynel-T3 was processed from LDPE-g-AM. The presence of MAH helps to prevent surface agglomeration and acts as a compatibilizer.<sup>46</sup> Thus, ZnO-NP dispersion was improved for Bynel-T3 samples, showing smaller aggregates. Moreover, using PE-g-AM has probably reinforced the ZnO-NP bonding to the Bynel-T3 film surface, possibly through hydrogen bonds. EDS analysis results, presented in Figure 3(c) and (d), show the EDS analysis which estimates the ZnO-NP surface concentration by using the mapping mode in the SEM-EDS equipment. The ZnO-NP concentration on the Bynel-T3 film surface was around 17 wt%, slightly higher than the FPs-T3 film which was around 14 wt%. These results are congruent with the hypothesis proposed above that the MAH attached more ZnO-NPs to the film surface and caused a good interaction between LDPE surface and ZnO-NPs.<sup>6</sup> It was also proved that in the case where a migratory antibacterial agent, the epigallocatechin gallate (EGCG), is used to develop active food packaging, the use of modified PE-resin can provide advantages to



**Figure 3.** Energy-dispersive X-ray spectroscopy (EDS-SEM) surface analysis of FP-T3 film (a, c) and Bynel-T3 film (b, d). The EDS samples were analyzed with a mapping mode operating at 15 kV.

active films. This study revealed that the good antibacterial performance of the EGCG-incorporated LDPE films against food-borne bacteria was due to the EGCG and the adhesive resin (LDPE with grafted functional groups). Moreover, it was demonstrated that the adhesive resin increased the EGCG release rate in the migration profile.<sup>47</sup>

To the best of our knowledge, it was the first time that this method was used to produce coated polymeric films with ZnO-NPs or with other AB metallic NPs. Otherwise, several methods have been used to produce ZnO coatings on different polymeric substrate. Thin ZnO and ZnO-Ag films were deposited on PET and silicon (Si) substrates by reactive DC magnetron sputtering.<sup>30</sup> In another study, ZnO thin films were applied to biaxially oriented poly(lactic) acid and polypropylene by atomic layer deposition at low temperatures.<sup>23</sup> In an earlier study, ZnO suspensions were coated on a PVC film surface. After drying, the film was pressed at 100°C for 10 min, and then it was peeled carefully after cooling at room temperature.<sup>18</sup> In the three studies listed above, the coated polymeric film ABA was evaluated by different methods. One interesting result is the one related to the PET substrate. That study found that the samples having thinner coatings

(50 and 100 nm) exhibited less ABA than the thicker coatings (200–600 nm), which were not only able to decrease the growth rate but also reduce the bacteria cells observed at the end of the experiments by 45%. This result is extremely useful for food packaging applications because it may reduce bacterial presence.<sup>30</sup>

### *Coated film long-term AB properties*

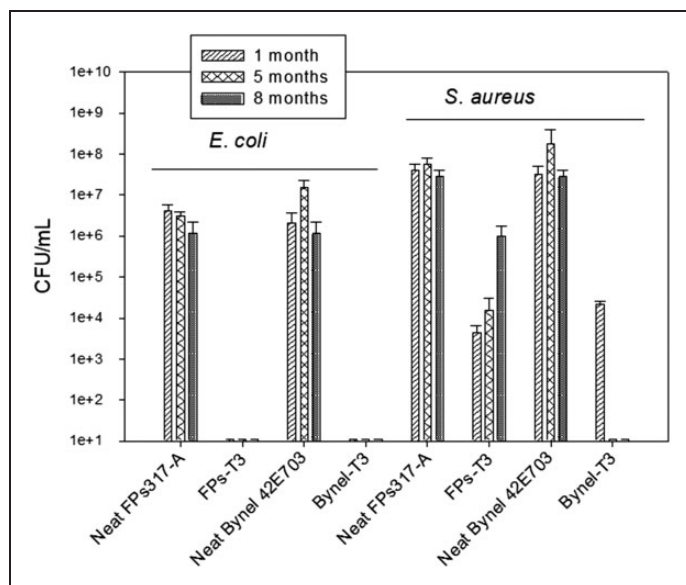
After their preparation, the FPs and Bynel films were stored at room temperature (21°C) and relative humidity between 7% and 40%, and their ABA was tested at five and eight months following the same steps to verify if their ABA will change with time. Figure 4 shows that after five and eight months, FPs-T3 and Bynel-T3 samples retained their AB properties, and the ZnO-NPs were able to decrease the *E. coli* population by more than 99.9999%.

For the Gram-positive strain (*S. aureus*), at one month after its preparation, the FPs-T3 films decreased the bacterial growth by approximately four logs (CFU/mL) to reach a population of  $4.3 \times 10^3$  CFU/mL. Four months later, the *S. aureus* population was around  $1.5 \times 10^4$  CFU/mL, but the inhibition was considerable since the control population was reduced by three logs (CFU/mL). At eight months after the film preparation, the FPs-T3 films reduced the bacterial population to  $9.8 \times 10^5$  CFU/mL, thus reducing the population by one log (CFU/mL). The slight loss of the AB activity of the FPs-T3 films between one and eight months can be attributed to a partial loss of the ZnO-NPs during storage because of handling and friction of the film on itself or on others surfaces. For the Bynel-T3 sample, the first test performed at one month post-preparation has revealed a reduction rate of three logs (CFU/mL), and then for the subsequent tests, the reduction rate exceeded 99.9%. This finding consolidated what was discussed above; ZnO-NPs were well impregnated on the PE-g-MA surface. Considering that ZnO-NPs can be lost by handling the samples, sufficient ZnO-NP concentration was still well attached to the surface of the film eight months after its preparation, thus reducing more than 99.9999% of the bacterial population.

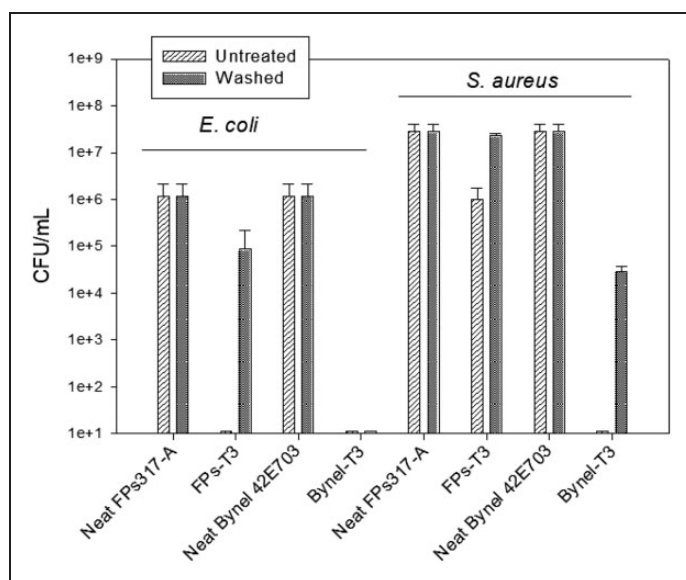
### *Coated film AB properties after ultrasound washing*

To confirm if the ZnO-NPs are well attached to LDPE films, the films were shaken in an ultrasonic bath for 20 min and then dried overnight in air at room temperature (21°C). During their preparation for the AB test, and before their inoculation, the films were sterilized by UV light for 20 min. Figure 5 reports the AB properties of the treated and untreated films.

The results show that for the Gram-negative bacteria, the washed FPs-T3 films reduced the *E. coli* growth by just one log (CFU/mL), whereas before washing the same film was able to kill all the bacterial population (more than 99.9%). By contrast, Bynel-T3 films still displayed a high ABA and decreased the bacterial density by more than 99.9% before and after washing.



**Figure 4.** Inhibitory effects of FPs-T3 and Bynel-T3 films toward *E. coli* and *S. aureus* at one, five, and eight months after their preparation.

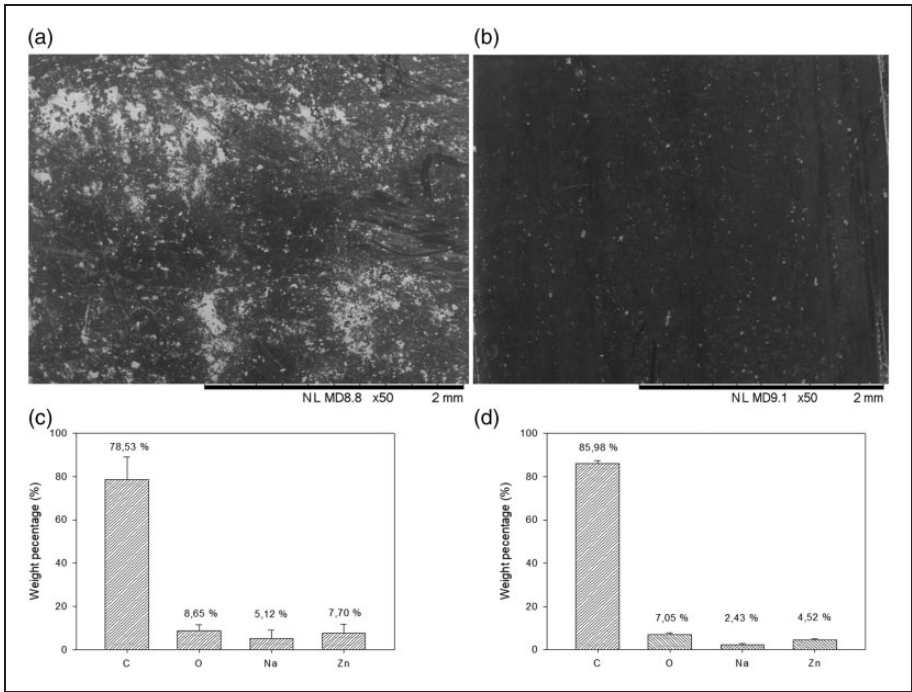


**Figure 5.** Inhibitory effects of treated and untreated FPs-T3 and Bynel-T3 films toward *E. coli* and *S. aureus*. The films were washed in an ultrasonic bath for 20 min and then dried in air.

For the *S. aureus* strain, the washed FPs-T3 films did not exhibit antibacterial properties when compared to the same film before washing that provided two logs (CFU/mL) reduction, while the washed Bynel-T3 films reduced the bacterial density by about three logs (CFU/mL). These films had lost much of their effectiveness since, before washing, they reduced the *S. aureus* density by approximately seven logs (CFU/mL).

Taking into account the previous results, it is imperative to determinate how much the amount of ZnO-NPs that remained on the surface after washing. EDS analysis, presented in Figure 6(c) and (d) estimate the ZnO-NP surface concentration by using the SEM-EDS mapping mode.

The ZnO-NP concentration on the washed FPs-T3 and Bynel-T3 film surface was around 7.7 and 4.5 wt%, respectively. The SEM images, shown in Figure 6(a) and (b), consolidated the results discussed previously saying that the ZnO-NP distribution on the film surface is more uniform for the Bynel-T3 films which was well illustrated in Figure 6(a) and (b). It is obvious that after washing, many ZnO-NPs have been removed. According to EDS results, the washed Bynel-T3 film surface was characterized by a ZnO-NP concentration of 4.5 wt%,



**Figure 6.** Energy-dispersive X-ray spectroscopy (EDS-SEM) surface analysis of FPs-T3 film (a, c) and Bynel-T3 film (b, d) surface after washing 20 min in an ultrasonic bath. EDS samples were analyzed with a mapping mode operating at 15 kV.

which is far below the concentration on the surface of washed FPs-T3 films, even though the ABA of the washed Bynel-T3 films was higher than that of the washed FPs-T3 films for both bacterial strains. In fact, the washed Bynel-T3 films decreased more than 99.9% of the *E. coli* population, whereas the washed FPs-T3 ones reduced the same bacterial population by only one log (CFU/mL). For *S. aureus*, the washed Bynel-T3 films reduced the bacterial density by about three logs (CFU/mL), while no ABA was recorded for the washed FPs-T3 films.

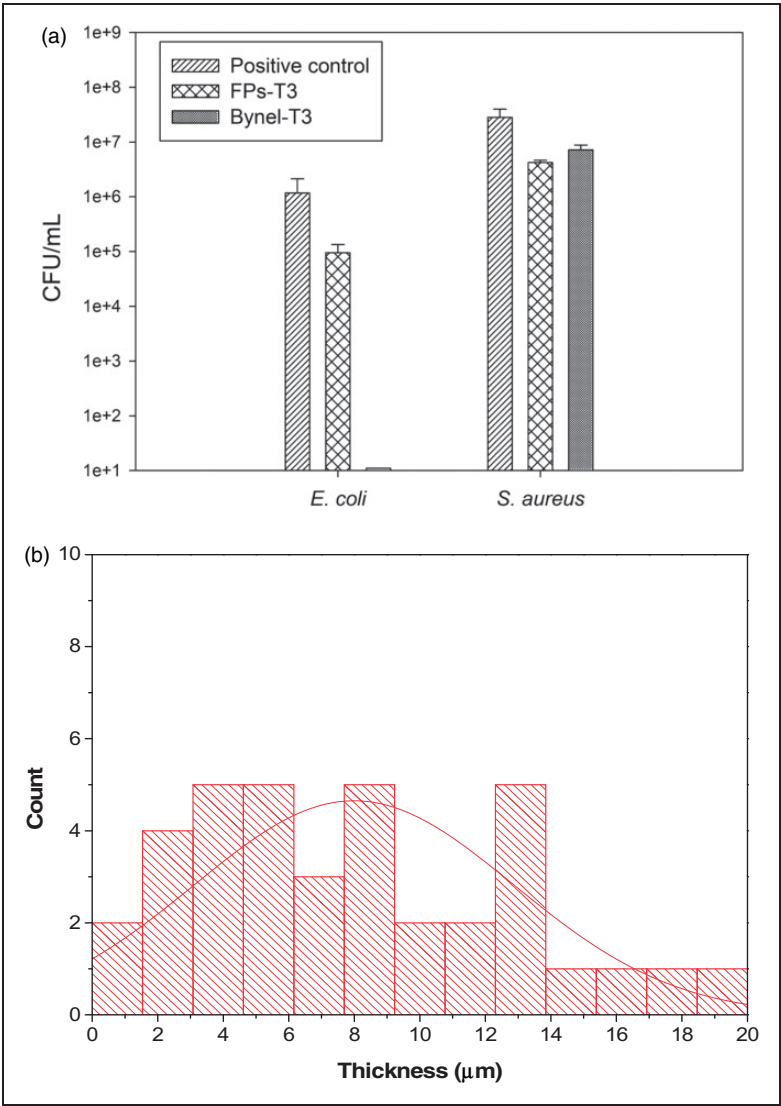
From the discussion above, one concludes that the ZnO-NP distribution has a large effect on the film's ABA. A lower concentration with better distribution was more efficient toward both bacterial strains. This effect was due to the PE-g-AM that increased the compatibility between the polymeric matrix and ZnO-NPs, allowing a better adhesion to the surface and smaller ZnO-NP agglomerates. Thereby, more surface area was available to cope with the bacteria. In fact, one parameter involved in ZnO-NP antibacterial properties is the specific surface area. It increases for reduced particle size, thus enhancing particle surface reactivity.<sup>14,15,42,48</sup> Moreover, generating H<sub>2</sub>O<sub>2</sub> is mainly dependent on the ZnO-NP surface area.<sup>42,49</sup> Padmavathy and Vijayaraghavan<sup>42</sup> studied the AMA of ZnO suspensions against *E. coli* with various particle sizes (12 nm, 45 nm, and 2  $\mu$ m) using a standard microbial method. They concluded that using small nanoparticles, thus having a larger surface area, will generate a higher ROS concentration on the surface and will result in greater ABA. More precisely, for smaller ZnO-NPs, more particles are needed to cover the bacterial colony (2  $\mu$ m), resulting in more ROS, killing bacteria more effectively.<sup>42</sup> By analogy, it can be assumed that smaller ZnO-NP agglomeration could provide more surface area and more ROS to ensure better ABA.

### *Comparative AB properties of the melt compounded films*

But if the ZnO-NPs will be blended with LDPE via the melt compounding, are they going to retain the same AB properties? To answer this question, a 20 wt% ZnO-NP master batch was prepared. Then, they were diluted to produce 5 wt% ZnO-NP cast films, the first one was processed from the Bynel resin and the second was an FPs and Bynel resin blend. The AB evaluation did not show any significant AB performance neither against Gram-positive nor against Gram-negative bacteria.

These results were predictable since the extrusion process as well as solvent casting are more suitable for migratory AB systems where the volatile AB agent release from the packaging material to the packaging headspace or the food product surface is required.

It is obvious that only when the ZnO-NPs are coated on the surface of the LDPE films and remain in direct contact with bacteria that they ensure efficient ABA against both Gram-positive and Gram-negative bacteria. The next section checks to see if the sprayed ZnO-NPs will retain their activity when they are laminated with another polymer layer or if their effectiveness will be lost.



**Figure 7.** (a) Inhibitory effects of different FPs and Bynel films toward *E. coli* and *S. aureus*. The films were coated by dipping in FPs317-A/xylene solution. (b) Coating layer thickness distribution in  $\mu\text{m}$ .

*Laminated films AB properties*

The laminated-LDPE films were prepared by the dipping method. A 2 wt% of FPs317-A resin xylene solution was prepared at 120°C. Once the resin is completely dissolved, the solution is cooled to 70°C, and the films were dipped for 1–2 s then

dried under the hood overnight. Figure 7(a) shows that the dipped FPs-T3 films reduced the *E. coli* population by just one log (CFU/mL), while the dipped Bynel-T3 was still efficient toward the bacteria and destroyed the bacterial density by more than 99.9%.

For *S. aureus*, the reduction was less than one log (CFU/mL) for both films. FPs-T3 films registered only one log (CFU/mL) reduction, while before lamination they were able to reduce the bacterial population by more than 99.9%. Figure 7(b) shows the coating layer thickness distribution and it shows that the average thickness was around 8  $\mu\text{m}$  for both dipped film sides. According to these results, one can assume that, for coating thicknesses less than 8  $\mu\text{m}$ , the coated Bynel-T3 could retain their activity toward *E. coli*. These films were characterized by a good ZnO-NP adhesion to the LDPE surface. To estimate the limit of this hypothesis, FPs-T3 film was laminated with a hot press with 18, 21, and 38  $\mu\text{m}$  PE films. Then, their ABA was evaluated. The results (not shown here) showed that there was no ABA toward *E. coli* and *S. aureus* compared to the control samples. It is obvious that when the lamination layer thickness is greater than 8  $\mu\text{m}$ , the ZnO-NPs will be too far away from the bacterial cells and lose their AB performance. To be efficient toward bacteria, ZnO-NPs should be in direct contact or close to the bacteria cells to ensure a good ABA.

## Conclusions

This study presented a new approach to coat LDPE films with metallic AB agents. Both LDPE and LDPE-g-AM coated films showed considerable inhibition against *E. coli* and *S. aureus*. For the FPs films, the ZnO-NP efficiency toward *S. aureus* was improved at higher processing temperature. SEM images of ZnO-coated LDPE films clearly showed that for LDPE-g-AM, the ZnO-NP distribution was more uniform, showing smaller aggregates. In this case, the compatibility between the polymeric matrix and the metal oxide NPs increased where the NPs were well attached to the film surface. The AB tests showed that the FPs film retained the same AB efficiency toward *E. coli* between one and eight months post-preparation. For *S. aureus*, the AB properties of the FPs-T3 film were slightly reduced, which may be due to a partial losing of the ZnO-NPs during storage or during sample preparation for the AB tests, since ZnO-NPs are not well attached to LDPE film surface. For the Bynel-T3 films, the ABA improved between one and eight months. When shaking in water, the FPs-T3 films lost their AB properties toward *E. coli* and *S. aureus* by contrast to Bynel-T3 films that were able to keep a reduction rate higher than three logs (CFU/mL). To maintain desired ABA, it is recommended that the Bynel-T3 should be laminated with a film layer whose thickness is less than 8  $\mu\text{m}$ . This coating approach guaranteed ZnO-NPs have a good ABA which was absent when NPs were compounded with the polymer in the melt state.

## Acknowledgements

Financial support of 3S Pack NSERC/Saputo/ProAmpac Industrial Research Chair from the Natural Sciences and Engineering Research Council of Canada and Industrial partners is gratefully acknowledged.

## Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study was funded by 3S Pack NSERC/Saputo/ProAmpac Industrial Research Chair from the Natural Sciences and Engineering Research Council of Canada and Industrial partners.

## ORCID iD

Hajer Rokbani  <http://orcid.org/0000-0001-9734-8991>

## References

1. Bruna J, et al. *Development of MtCu 2+/LDPE nanocomposites with antimicrobial activity for potential use in food packaging*. *Appl Clay Sci* 2012; 58: 79–87.
2. Rhim J-W, Park H-M and Ha C-S. Bio-nanocomposites for food packaging applications. *Prog Polym Sci* 2013; 38: 1629–1652.
3. Abdali H and Ajjai A. Development of antibacterial structures and films using clove bud powder. *Ind Crops Prod* 2015; 72: 214–219.
4. Garlotta D. A literature review of poly (lactic acid). *J Polym Environ* 2001; 9: 63–84.
5. Del Nobile M, et al. Active packaging by extrusion processing of recyclable and biodegradable polymers. *J Food Eng* 2009; 93: 1–6.
6. Emamifar A and Mohammadizadeh M. Preparation and application of LDPE/ZnO nanocomposites for extending shelf life of fresh strawberries. *Food Technol Biotechnol* 2015; 53: 488–495.
7. Sawai J. Quantitative evaluation of antibacterial activities of metallic oxide powders (ZnO, MgO and CaO) by conductimetric assay. *J Microbiol Methods* 2003; 54: 177–182.
8. Sawai J and Yoshikawa T. Quantitative evaluation of antifungal activity of metallic oxide powders (MgO, CaO and ZnO) by an indirect conductimetric assay. *J Appl Microbiol* 2004; 96: 803–809.
9. Ng AMC, et al. Antibacterial and photocatalytic activity of TiO<sub>2</sub> and ZnO nanomaterials in phosphate buffer and saline solution. *Appl Microbiol Biotechnol* 2013; 97: 5565–5573.
10. De Azeredo HM. Nanocomposites for food packaging applications. *Food Res Int* 2009; 42: 1240–1253.
11. Bumbudsanpharoke N and Ko S. Nano-food packaging: an overview of market, migration research, and safety regulations. *J Food Sci* 2015; 80: 910–923.

12. Jones N, et al. Antibacterial activity of ZnO nanoparticle suspensions on a broad spectrum of microorganisms. *FEMS Microbiol Lett* 2008; 279: 71–76.
13. Shi L-E, et al. Synthesis, antibacterial activity, antibacterial mechanism and food applications of ZnO nanoparticles: a review. *Food Addit Contam Part A* 2014; 31: 173–186.
14. Zhang L, et al. Investigation into the antibacterial behaviour of suspensions of ZnO nanoparticles (ZnO nanofluids). *J Nanopart Res* 2007; 9: 479–489.
15. Yamamoto O. Influence of particle size on the antibacterial activity of zinc oxide. *Int J Inorg Mater* 2001; 3: 643–646.
16. Stoimenov PK, et al. Metal oxide nanoparticles as bactericidal agents. *Langmuir* 2002; 18: 6679–6686.
17. Dutta RK, et al. Differential susceptibility of Escherichia coli cells toward transition metal-doped and matrix-embedded ZnO nanoparticles. *J Phys Chem B* 2010; 114: 5594–5599.
18. Li X, et al. Antibacterial and physical properties of poly(vinyl chloride)-based film coated with ZnO nanoparticles. *Food Sci Technol Int* 2010; 16: 225–232.
19. Espitia PJP, et al. Zinc oxide nanoparticles: synthesis, antimicrobial activity and food packaging applications. *Food Bioprocess Technol* 2012; 5: 1447–1464.
20. Petersen K, et al. Potential of biobased materials for food packaging. *Trends Food Sci Technol* 1999; 10: 52–68.
21. Arvanitoyannis I, et al. Biodegradable films made from low density polyethylene (LDPE), ethylene acrylic acid (EAA), polycaprolactone (PCL) and wheat starch for food packaging applications: Part 3. *Starch/Stärke* 1997; 49: 306–322.
22. Pantani R, et al. PLA-ZnO nanocomposite films: Water vapor barrier properties and specific end-use characteristics. *Eur Polym J* 2013; 49: 3471–3482.
23. Vähä-Nissi M, et al. Antibacterial and barrier properties of oriented polymer films with ZnO thin films applied with atomic layer deposition at low temperatures. *Thin Solid Films* 2014; 562: 331–337.
24. Appelerot G, et al. Coating of glass with ZnO via ultrasonic irradiation and a study of its antibacterial properties. *Appl Surf Sci* 2009; 256: S3–S8.
25. Emamifar A, et al. Evaluation of nanocomposite packaging containing Ag and ZnO on shelf life of fresh orange juice. *Innov Food Sci Emerg Technol* 2010; 11: 742–748.
26. Emamifar A, et al. Effect of nanocomposite packaging containing Ag and ZnO on inactivation of *Lactobacillus plantarum* in orange juice. *Food Control* 2011; 22: 408–413.
27. Lepot N, et al. Influence of incorporation of ZnO nanoparticles and biaxial orientation on mechanical and oxygen barrier properties of polypropylene films for food packaging applications. *J Appl Polym Sci* 2011; 120: 1616–1623.
28. Li J, et al. Effects of ZnO nanoparticles on the mechanical and antibacterial properties of polyurethane coatings. *Prog Org Coat* 2009; 64: 504–509.
29. Petchwattana N, et al. Antimicrobial food packaging prepared from poly (butylene succinate) and zinc oxide. *Measurement* 2016; 93: 442–448.
30. Carvalho P, et al. Influence of thickness and coatings morphology in the antimicrobial performance of zinc oxide coatings. *Appl Surf Sci* 2014; 307: 548–557.
31. Ghule K, et al. Preparation and characterization of ZnO nanoparticles coated paper and its antibacterial activity study. *Green Chem* 2006; 8: 1034–1041.
32. Wang Y, et al. Characterisation and cooperative antimicrobial properties of chitosan/nano-ZnO composite nanofibrous membranes. *Food Chem* 2012; 132: 419–427.

33. Vicentini DS, Smania A and Laranjeira MC. Chitosan/poly (vinyl alcohol) films containing ZnO nanoparticles and plasticizers. *Mater Sci Eng C* 2010; 30: 503–508.
34. Tawakkal IS, et al. A review of poly (lactic acid)-based materials for antimicrobial packaging. *J Food Sci* 2014; 79: R1477–R1490.
35. Jalal R, et al. ZnO nanofluids: green synthesis, characterization, and antibacterial activity. *Mater Chem Phys* 2010; 121: 198–201.
36. Sawai J, et al. Hydrogen peroxide as an antibacterial factor in zinc oxide powder slurry. *J Ferment Bioeng* 1998; 86: 521–522.
37. Lipovsky A, et al. Antifungal activity of ZnO nanoparticles—the role of ROS mediated cell injury. *Nanotechnology* 2011; 22: 105101.
38. Brayner R, et al. Toxicological impact studies based on *Escherichia coli* bacteria in ultrafine ZnO nanoparticles colloidal medium. *Nano Lett* 2006; 6: 866–870.
39. Adams LK, Lyon DY and Alvarez PJ. Comparative eco-toxicity of nanoscale TiO<sub>2</sub>, SiO<sub>2</sub>, and ZnO water suspensions. *Water Res* 2006; 40: 3527–3532.
40. Zhang L, et al. ZnO nanofluids – a potential antibacterial agent. *Prog Nat Sci* 2008; 18: 939–944.
41. Reddy KM, et al. Selective toxicity of zinc oxide nanoparticles to prokaryotic and eukaryotic systems. *Appl Phys Lett* 2007; 90: 213902.
42. Padmavathy N and Vijayaraghavan R. Enhanced bioactivity of ZnO nanoparticles – an antimicrobial study. *Sci Technol Adv Mater* 2008; 9: 035004.
43. Xie Y, et al. Antibacterial activity and mechanism of action of zinc oxide nanoparticles against *Campylobacter jejuni*. *Appl Environ Microbiol* 2011; 77: 2325–2331.
44. Apperlot G, et al. Enhanced antibacterial activity of nanocrystalline ZnO due to increased ROS-mediated cell injury. *Adv Funct Mater* 2009; 19: 842–852.
45. ISO22196. *Measurement of antibacterial activity on plastics and other non-porous surfaces*. London, UK: International Organization for Standardization, 2011, p.15.
46. Majid RA, Ismail H and Taib RM. Effects of polyethylene-g-maleic anhydride on properties of low density polyethylene/thermoplastic sago starch reinforced kenaf fibre composites. *Iran Polym J* 2010; 19: 501–510.
47. Moreno-Vásquez MJ, et al. Engineering and antibacterial properties of low-density polyethylene films with incorporated epigallocatechin gallate. *J Plast Film Sheeting* 2017; 33: 413–437.
48. Zhang L, et al. Mechanistic investigation into antibacterial behaviour of suspensions of ZnO nanoparticles against *E. coli*. *J Nanopart Res* 2010; 12: 1625–1636.
49. Ohira T and Yamamoto O. Correlation between antibacterial activity and crystallite size on ceramics. *Chem Eng Sci* 2012; 68: 355–361.

## Biographies

**Hajer Rokbani** has a PhD in Chemical Engineering from “École Polytechnique de Montréal” Quebec, Canada, with a strong background in plastics and antibacterial food packaging. She joined NSERC-Industry research Chair on Materials and Film for Safe, Smart and Sustainable Packaging (3SPack) in 2012. She obtained her engineering degree from “Institut National des Sciences Appliquées et de Technologie” in Tunis, Tunisia, in 2009 and her master’s degree in chemistry from the same school in 2011. Her academic research explores the possible strategy to develop smart food packaging with variable properties. During these years of

experience, she developed her skills in polymer processing, polymer's characterization, nanopowder's characterization and biomaterials.

**France Daigle** is a Professor in the Department of Microbiology, Infectiology and Immunology at the "Université de Montréal". She is a bacteriologist and her projects involve the study of molecular pathogenesis of Salmonella. She has expertise in host-pathogen interactions, using host cells as a model to study adherence, uptake, and survival and studies bacterial gene expression. She is the Director of the BSc program in microbiology and immunology.

**Abdellah Aji** is a Professor of Chemical Engineering at "École Polytechnique de Montréal" Quebec, Canada, and Holder of an NSERC-Industry research Chair on Materials and Film for Safe, Smart and Sustainable Packaging (3SPack). He obtained his engineering degree from "École Nationale Supérieure des Industries Chimiques" in Nancy, France, and his MSc and PhD in Chemical Engineering from "École Polytechnique de Montréal". He is highly involved in research with main interests in correlations between the process, structure, and properties for polymer systems in the form of films and fibers, particularly multilayer and functional systems. He published in this area of research and others with more than 180 articles in peer reviewed scientific journals, many book chapters, patents, and patent applications.